BIO 280 PATHOPHYSIOLOGY · JOYCE UNIVERSITY

Dr. Garg's Exam 5 Master Study Guide

Modules 9 & 10 — Blood Disorders & the Cardiovascular System

ⓘ What Exam 5 covers

This unit is about **the blood and the heart**. **Module 9 (Blood Disorders)** covers the anemias, B12/pernicious anemia, thalassemia, polycythemia vera, the bleeding/clotting disorders (von Willebrand, DIC), and transfusion reactions. **Module 10 (Cardiovascular)** covers coronary artery disease and MI, heart failure, hypertension, arrhythmias, the inflammatory and valvular heart diseases, peripheral vascular disease, and congenital defects. Exam format: **50 questions (multiple choice + select-all-that-apply)**, proctored with Respondus LockDown Browser.

ⓘ How this guide works

- **Click any topic card** in the navigator to jump to that section. Click a title bar to expand or close it.
- **All answers are hidden** — try first, then click "Show answer." Practice includes **multiple choice** and **select-all-that-apply (SATA)**, like the real exam.
- **Each topic has:** clear explanation, a mnemonic, a "common confusion" warning, practice questions, and a clinical scenario.
- **Color code:** **crimson = Module 9 (Blood)**, **blue = Module 10 (Cardiovascular)**.
- **To save as PDF:** use the button above, or press *Ctrl+P* / *Cmd+P* → "Save as PDF." All sections expand for printing.

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TOPIC 1

□Anemias: Overview & Types

MUST KNOW ▼

Anemia = reduced oxygen-carrying capacity (low hemoglobin/RBCs). Whatever the cause, the **general signs/symptoms** are the same: **fatigue, pallor, tachycardia, dyspnea on exertion, dizziness, cold intolerance** — all from tissue hypoxia.

Classify by red-cell size (MCV) — the fastest first cut

Type	Patho / cause	Hallmark clue
Iron-deficiency (microcytic)	Too little iron → ↓ hemoglobin (blood loss, poor intake, pregnancy)	↓ **ferritin**, spoon nails (koilonychia), pica
Thalassemia (microcytic)	**Inherited** defect → reduced globin-chain production → abnormal/fragile RBCs	**Normal iron studies**, target cells, family history
B12 / folate (macrocytic)	Impaired **DNA synthesis** → large, immature RBCs	**B12 → neuro signs** (paresthesias, ataxia); folate does not
Sickle cell (normo, hemolytic)	Point mutation → HbS **sickles when deoxygenated** → vaso-occlusion + hemolysis	Pain crises; triggers: hypoxia, dehydration, infection, cold
Aplastic (normo)	**Bone-marrow failure → pancytopenia** (drugs, radiation, idiopathic)	All lines low → anemia + infection + bleeding; **low** reticulocytes
Hemolytic (normo)	RBCs destroyed faster than made	↑ **reticulocytes,** ↑ **indirect bilirubin,** ↑ **LDH,** ↓ **haptoglobin**; jaundice

B12 deficiency & pernicious anemia

Pernicious anemia is the classic B12 cause: an **autoimmune loss of intrinsic factor** (made by gastric parietal cells), so B12 can't be absorbed. Signs: macrocytic anemia + **neurologic** symptoms (numbness/tingling, balance problems), glossitis, and fatigue.

□Memory hooks

Microcytic = "small & iron-poor" (iron deficiency, thalassemia). **Macrocytic** = "big &

DNA-broken" (B12/folate). **Only B12 gives neuro signs.** **Aplastic = "absent marrow"** → everything low. Pernicious = **"no intrinsic factor."**

⚠ **Common confusion**

Iron deficiency vs thalassemia (both microcytic): iron deficiency has **low ferritin**; thalassemia has **normal iron studies** and a family history. **B12 vs folate** (both macrocytic): only **B12** causes neurologic symptoms.

Practice Questions

1. A patient has microcytic anemia with low ferritin, brittle spoon-shaped nails, and cravings for ice. Which anemia?

   Show answer

   Iron-deficiency anemia. Low ferritin (empty stores), koilonychia (spoon nails), and pica are classic.

2. A vegan patient develops macrocytic anemia plus numbness and an unsteady gait. B12 or folate deficiency, and why?

   Show answer

   B12 deficiency — only B12 (needed for myelin) causes the neurologic signs. Folate deficiency gives the same big cells *without* neuro involvement.

3. **SATA:** Which are general signs/symptoms of anemia? (a) fatigue, (b) pallor, (c) tachycardia, (d) hypertension crisis, (e) dyspnea on exertion.

   Show answer

   a, b, c, e. Anemia causes fatigue, pallor, tachycardia (compensatory), and exertional dyspnea. A hypertensive crisis is not an anemia feature.

4. A patient has pancytopenia (low RBCs, WBCs, and platelets) with a low reticulocyte count after chemotherapy. Which anemia?

   Show answer

   Aplastic anemia — bone-marrow failure drops all three cell lines; the marrow can't respond, so reticulocytes are low.

Clinical scenario

A child with sickle cell disease is admitted with severe bone pain after playing outside in cold weather while dehydrated. **What is happening and why?**

*A **vaso-occlusive crisis**: cold, dehydration, and hypoxia caused HbS to **polymerize/sickle**, and the rigid cells block small vessels → ischemia and pain. Management targets the triggers — hydration, oxygen, warmth, and pain control.*

TOPIC 2

Bleeding, Clotting & Transfusion

MUST KNOW ▼

Module 9's other half: too *little* clotting (bleeding disorders), too *much*/consumed clotting (DIC), too many cells (polycythemia vera), and the dangerous transfusion reaction.

Disorder	Patho
Von Willebrand disease	Deficient/defective **von Willebrand factor** → platelets can't adhere well → **mucosal bleeding** (nosebleeds, easy bruising, heavy menses), prolonged bleeding time. Most common inherited bleeding disorder.
Disseminated intravascular coagulation (DIC)	**Widespread clotting** consumes platelets and clotting factors → then **severe bleeding**. Triggered by sepsis, trauma, obstetric events, malignancy. Labs: ↑**PT,** ↑**aPTT,** ↓**platelets,** ↓**fibrinogen,** ↑↑**D-dimer**.
Polycythemia vera	**JAK2** mutation → marrow overproduces RBCs (± WBCs/platelets) → thick, viscous blood. S/S: **ruddy (red) complexion, headache, pruritus after a warm shower, hypertension, thrombosis risk**.
Acute hemolytic transfusion reaction	**ABO incompatibility** → recipient antibodies attack donor RBCs → rapid **intravascular hemolysis** within minutes.

Acute hemolytic reaction — why it happens & what you see

If mismatched (ABO-incompatible) blood is given, the recipient's preformed antibodies bind donor red cells and destroy them. You see **fever, chills, flank/back pain, dark urine (hemoglobinuria), hypotension** within minutes — it can progress to shock, DIC, and acute kidney injury. **First action: STOP the transfusion** and keep the line open with saline.

Memory hooks

vWF = "the platelet glue" (its loss = mucosal bleeding). **DIC = "clots then bleeds"** (everything consumed; D-dimer sky-high). **Polycythemia vera = "too much, too thick, too red"** → clot risk. Acute hemolytic reaction = **ABO mismatch** → STOP + saline.

⚠ **Common confusion**

Polycythemia vera is too *many* cells (thrombosis risk); the **anemias** are too *few*. And in **DIC** the paradox is clotting *and* bleeding at once — not one or the other.

Practice Questions

1. Minutes into a transfusion, a patient develops fever, chills, flank pain, and dark urine. What is happening and what is the priority action?

   Show answer

   An **acute hemolytic transfusion reaction** (ABO incompatibility) — recipient antibodies destroy donor RBCs. **Stop the transfusion immediately** and keep the line open with normal saline; notify the provider/blood bank.

2. A patient with sepsis is bleeding from IV sites with ↑PT, ↑aPTT, low platelets, low fibrinogen, and very high D-dimer. Which disorder?

   Show answer

   DIC. Widespread clotting consumes platelets and factors (so PT/aPTT rise, platelets/fibrinogen fall) while fibrin breakdown drives the very high D-dimer — the patient clots and bleeds at once.

3. A patient has a ruddy complexion, headaches, itching after warm showers, and a JAK2 mutation. Which disorder, and what is the main risk?

   Show answer

   Polycythemia vera. The thickened, hyperviscous blood raises the risk of **thrombosis** (clots) — treated with phlebotomy and low-dose aspirin.

Clinical scenario

A teenager reports lifelong easy bruising, frequent nosebleeds, and heavy periods, with a normal platelet count but a prolonged bleeding time. **What is the likely disorder and mechanism?**

Von Willebrand disease *— deficient/defective vWF impairs platelet adhesion (and destabilizes factor VIII), producing mucosal-type bleeding with a normal platelet count but prolonged bleeding time.*

TOPIC 3

□Atherosclerosis & Coronary Artery Disease

MUST KNOW

Atherosclerosis is the buildup of lipid plaque in artery walls — the root of coronary artery disease (CAD), MI, stroke, and peripheral artery disease.

Stages of advancement

1. **Endothelial injury** (from hypertension, smoking, high LDL, diabetes) — the starting point.
2. **Fatty streak** — LDL enters the wall, is oxidized, and macrophages engulf it → **foam cells**.
3. **Fibrous plaque** — smooth muscle and collagen form a cap over the lipid core; the lumen narrows.
4. **Complicated (unstable) plaque** — the cap **ruptures** → a clot (thrombus) forms → acute occlusion (MI/stroke).

Risk factors

Modifiable	Non-modifiable
Smoking, hypertension, high LDL/low HDL, diabetes, obesity, sedentary lifestyle, poor diet, stress	**Age, male sex, family history/genetics**

□Memory hook

Progression: **Injury → Fatty streak (foam cells) → Fibrous plaque → Rupture & clot.** The **rupture** is what turns stable disease into an acute MI — a stable narrowing suddenly becomes a total blockage.

⚠ Common confusion

A **stable** plaque narrows the lumen (causes stable angina with exertion). An **unstable** plaque that **ruptures** triggers a clot and causes acute coronary syndrome (unstable angina/MI) — the danger is rupture, not just size.

□Practice Questions

1. Which event most directly converts stable atherosclerotic disease into an acute myocardial infarction?

Show answer

Plaque rupture with thrombus (clot) formation — the exposed lipid core triggers a clot that suddenly occludes the artery.

2. Which cells form the fatty streak, the earliest visible atherosclerotic lesion?

   Show answer

   Foam cells — macrophages that have engulfed oxidized LDL within the injured vessel wall.

3. **SATA:** Which are modifiable risk factors for CAD? (a) smoking, (b) hypertension, (c) family history, (d) diabetes, (e) age.

   Show answer

   a, b, d. Smoking, hypertension, and diabetes are modifiable. Family history (c) and age (e) are non-modifiable.

Clinical scenario

A 58-year-old smoker with hypertension and diabetes has chest tightness when climbing stairs that resolves with rest. **What stage/type of disease does this suggest?**

*A **stable fibrous plaque** narrowing a coronary artery → **stable angina**: demand-related chest pain with exertion that resolves with rest. His modifiable risk factors (smoking, HTN, diabetes) are driving the atherosclerosis.*

TOPIC 4

□Angina, MI & Troponin

MUST KNOW ▼

When myocardial oxygen **demand exceeds supply**, the heart becomes ischemic. The spectrum runs from angina to full infarction.

Condition	Patho	Key features
Stable angina	Fixed narrowing; ischemia only when **demand rises** (exertion)	Predictable pain, **relieved by rest/ nitroglycerin**; no cell death
Unstable angina	Plaque rupture/partial clot; ischemia **at rest**	New/worsening pain at rest; **normal troponin** (no infarct yet) — a warning sign
NSTEMI	**Partial** coronary occlusion → **subendocardial** infarct	↑ **Troponin**, ST **depression**/T-wave changes (no ST elevation)
STEMI	**Complete** coronary occlusion → **full-thickness** infarct	↑ **Troponin + ST elevation** — a true emergency (reperfuse fast)

Troponin & MI symptoms

Troponin is a protein released when **heart muscle is damaged**, so it is **elevated in MI** (NSTEMI/STEMI) and other myocardial injury (e.g., myocarditis). It is the key lab to distinguish **unstable angina (normal troponin)** from an **infarct (elevated troponin)**. Classic MI symptoms: **crushing substernal chest pain** (may radiate to left arm/jaw), diaphoresis, nausea, dyspnea; women/elderly/diabetics may have atypical presentations.

Sudden cardiac death

The most common cause of **sudden cardiac death** is a lethal **ventricular arrhythmia (ventricular fibrillation)**, usually triggered by underlying **coronary artery disease/ ischemia**.

□Memory hooks

Stable = with exertion, relieved by rest. Unstable = at rest, troponin normal. NSTEMI = troponin up, ST down/partial. STEMI = troponin up + ST elevation, total occlusion. "STEMI = ST-Elevation = Emergency."

⚠ **Common confusion**

The line between **unstable angina and NSTEMI** is **troponin**: both can have rest pain and no ST elevation, but only NSTEMI has **positive troponin** (actual cell death).

Practice Questions

1. A patient has chest pain at rest, no ST elevation, and a normal troponin. Which diagnosis?

   Show answer

   Unstable angina — rest pain from a ruptured/partially occluding plaque, but troponin is normal because no infarction has occurred yet. It's a warning that an MI may follow.

2. A patient has crushing chest pain, ST elevation on ECG, and elevated troponin. Which diagnosis and how urgent?

   Show answer

   STEMI — a complete coronary occlusion causing full-thickness infarction. It's a time-critical emergency requiring immediate reperfusion (PCI or thrombolytics).

3. When is troponin elevated?

   Show answer

   When there is **myocardial cell damage** — classically MI (NSTEMI/STEMI), and also in myocarditis or other cardiac injury. It stays normal in stable and unstable angina (no infarct).

Clinical scenario

A 62-year-old collapses; monitor shows ventricular fibrillation. **What is the most common underlying cause of sudden cardiac death here?**

Coronary artery disease/ischemia *triggering a lethal ventricular arrhythmia (VF). Immediate defibrillation and CPR are the priority.*

TOPIC 5

□Heart Failure & Preload/Afterload

MUST KNOW ▼

Heart failure = the heart can't pump enough to meet the body's needs. Split it two ways: **which side** and **systolic vs diastolic**.

	Systolic HF (HFrEF)	Diastolic HF (HFpEF)
Problem	Weak pump — ↓ **contractility** → ↓ ejection fraction (<40%)	Stiff ventricle — **can't relax/fill**; ejection fraction preserved (>50%)
Causes	MI, dilated cardiomyopathy, chronic ischemia, valve regurgitation	Long-standing hypertension with LVH, hypertrophic/restrictive cardiomyopathy, aging

	Left-sided HF	Right-sided HF
Backs up into	the **lungs**	the **systemic veins/body**
Signs	**Dyspnea, orthopnea, crackles, pink frothy sputum, pulmonary edema**	**JVD, peripheral edema, hepatomegaly, ascites, weight gain**
Common cause	HTN, ischemic/valve disease	**Most often left HF**; also lung disease (cor pulmonale)

Preload, afterload & the natriuretic peptides

- **Preload** = the volume/stretch in the ventricle before it contracts. ↑ by fluid overload, valve regurgitation; ↓ by dehydration/hemorrhage.
- **Afterload** = the resistance the ventricle pumps against. ↑ by **hypertension** and aortic stenosis (the heart works harder).
- **ANP/BNP** are released by **stretched heart chambers** (volume overload). They promote **sodium and water excretion + vasodilation** to lower volume/pressure — the body's own diuretic. **BNP is elevated in heart failure** and used to diagnose it.

□Memory hooks

Left = Lungs (pulmonary congestion). **Right = Rest of the body** (JVD, edema).
Systolic = squeeze problem (↓EF); **diastolic = fill problem** (stiff, EF preserved).
BNP up = the heart is stretched/failing.

> ⚠ **Common confusion**
> **Preload vs afterload:** preload = volume coming *in* (stretch before contraction); afterload = resistance pumping *out* (raised by **hypertension**). And the #1 cause of **right** HF is **left** HF.

Practice Questions

1. A patient has dyspnea, orthopnea, crackles, and pink frothy sputum. Left or right heart failure, and why?

   Show answer

   Left-sided HF — blood backs up into the lungs (pulmonary congestion) → dyspnea, crackles, and pulmonary edema.

2. A patient has jugular venous distension, peripheral edema, and hepatomegaly. Which side of the heart is failing?

   Show answer

   Right-sided HF — blood backs up into the systemic veins → JVD, dependent edema, and liver congestion.

3. Which condition most increases afterload on the left ventricle?

   Show answer

   Hypertension (and aortic stenosis) — the LV must generate more force to eject against the higher resistance.

4. Why is BNP elevated in heart failure?

   Show answer

   Volume overload **stretches the heart chambers**, which release BNP to promote sodium/water excretion and vasodilation. A high BNP supports a heart-failure diagnosis.

Clinical scenario

A patient with long-standing hypertension has exertional dyspnea; echo shows a **preserved ejection fraction** but a thick, stiff left ventricle. **Which type of heart failure?**

Diastolic heart failure (HFpEF) *— chronic hypertension caused LV hypertrophy; the stiff ventricle can't relax and fill well, even though the squeeze (EF) is preserved.*

TOPIC 6

□Hypertension & Blood-Loss Compensation

MUST KNOW

Hypertension is chronically elevated arterial pressure. It's usually **"silent"** (few symptoms) until it damages organs — which is why it's the "silent killer."

Contributing factors & symptoms

- **Contributing factors:** age, obesity, high sodium intake, sedentary lifestyle, excess alcohol, smoking, stress, family history, diabetes, kidney disease (most primary/essential HTN has no single cause).
- **Common symptoms:** often **none**; when present — headache, dizziness, blurred vision; severe elevations can cause chest pain or epistaxis.

Complications (target-organ damage)

Organ	Damage
Heart	LV hypertrophy → heart failure; accelerates CAD/MI
Brain	**Stroke**, aneurysm
Kidneys	Chronic kidney disease (damages small renal vessels)
Eyes / vessels	Retinopathy; aneurysm/dissection; peripheral artery disease

Compensatory mechanisms during blood loss

When blood volume drops (hemorrhage), the body defends blood pressure in a set order:

- **Fast (seconds):** baroreceptors trigger **sympathetic activation → tachycardia, increased contractility, vasoconstriction**.
- **Minutes–hours: RAAS** (renin → angiotensin II vasoconstriction + aldosterone Na/water retention) and **ADH** (water retention) restore volume.
- **Slow (days):** the kidney releases **erythropoietin** → more RBCs; fluid shifts from interstitium into vessels.

□Memory hooks

HTN = the **"silent killer"** that damages the **heart, brain, kidneys, and eyes**. Blood-

loss compensation order: **Sympathetic (fast) → RAAS/ADH (volume) → EPO (slow).**

⚠ **Common confusion**

Hypertension is usually **asymptomatic** — don't wait for symptoms. The danger is silent **target-organ damage** (stroke, HF, CKD) over years.

Practice Questions

1. **SATA:** Which are complications of chronic hypertension? (a) stroke, (b) heart failure/LVH, (c) chronic kidney disease, (d) improved vision, (e) MI.

   Show answer

   a, b, c, e. HTN damages the brain (stroke), heart (LVH/HF, MI), and kidneys (CKD). It harms, not improves, vision (retinopathy).

2. Which is the FIRST compensatory response to acute blood loss?

   Show answer

   Baroreceptor-mediated sympathetic activation — immediate tachycardia, increased contractility, and vasoconstriction to defend blood pressure (RAAS/ADH and EPO follow more slowly).

3. Why is hypertension called the "silent killer"?

   Show answer

   It's usually **asymptomatic** while quietly causing **target-organ damage** (heart, brain, kidney, eyes) over time.

Clinical scenario

A trauma patient loses a large volume of blood and becomes tachycardic with cool, pale skin and a narrow pulse pressure. **What compensation explains these findings?**

*Rapid **sympathetic activation** from baroreceptors: tachycardia (to maintain output) and peripheral **vasoconstriction** (cool, pale skin) defend blood pressure while RAAS/ADH work to restore volume.*

TOPIC 7

⚡ Arrhythmias & Atrial Fibrillation

MUST KNOW ▼

An **arrhythmia** is an abnormal heart rhythm from disordered electrical activity. The exam favorite is **atrial fibrillation**.

Atrial fibrillation (AFib)

- **Patho:** chaotic, disorganized atrial electrical activity → the atria **quiver instead of contracting** → irregularly irregular rhythm, loss of the coordinated "atrial kick."
- **Biggest complication:** blood **pools/stagnates** in the fibrillating atria → clots form → a clot can embolize → **stroke** (which is why AFib patients are anticoagulated). Can also cause ↓ cardiac output and heart failure.

Arrhythmia complications in general

Depending on rate/rhythm: **decreased cardiac output** (dizziness, syncope, hypotension), **thromboembolism/stroke** (AFib), **heart failure**, and — with lethal ventricular rhythms (VT/VF) — **sudden cardiac death**.

Memory hook

AFib = "quivering atria → pooled blood → clot → stroke." That embolic-stroke risk is why AFib patients get anticoagulation. "Irregularly irregular" pulse = AFib.

⚠ Common confusion

Atrial fibrillation (AFib) → stroke risk, often chronic/manageable. **Ventricular** fibrillation (VF) → no effective pumping → **cardiac arrest** requiring immediate defibrillation. Don't mix the two.

Practice Questions

1. Which mechanism explains the increased stroke risk in atrial fibrillation?

   Show answer

   The atria quiver instead of contracting, so blood **stagnates and clots**; a clot can embolize to the brain → stroke. This is why AFib patients are anticoagulated.

2. **SATA:** Which are potential complications of arrhythmias? (a) decreased cardiac output, (b) thromboembolic stroke, (c) heart failure, (d) sudden cardiac death, (e) improved perfusion.

   Show answer

   a, b, c, d. Arrhythmias can drop cardiac output, throw clots (AFib), precipitate HF, and cause sudden death (VT/VF). They don't improve perfusion.

Clinical scenario

An older adult with a new "irregularly irregular" pulse suddenly develops right-sided weakness and slurred speech. **What likely happened?**

*New-onset **atrial fibrillation** allowed a clot to form in the stagnant atrium; it embolized to the brain → an **embolic stroke**. This is exactly why AFib warrants anticoagulation.*

TOPIC 8

□Inflammatory & Valvular Heart Disease

MUST KNOW ▼

Infections and inflammation can strike each layer of the heart, and the valves. Match the layer to the disease.

Condition	Patho / signs
Pericarditis	Inflammation of the **pericardium** → sharp, **pleuritic chest pain relieved by sitting forward**, a **friction rub**
Pericardial effusion	Fluid accumulates in the pericardial sac; may cause muffled heart sounds, dyspnea
Cardiac tamponade	**Rapid/large effusion compresses the heart** → can't fill → ↓ cardiac output. **Beck's triad: hypotension, JVD, muffled heart sounds** — emergency (pericardiocentesis)
Myocarditis	Inflammation of the **myocardium** (often viral) → weakened pump, arrhythmias, HF; may ↑ troponin
Infective endocarditis	Infection of the **endocardium/valves** → **vegetations**. Fever, new murmur, **Janeway lesions, Osler nodes, splinter hemorrhages**; risk with IV drug use, prosthetic valves
Rheumatic heart disease	Autoimmune reaction **after untreated strep** (rheumatic fever) → scars heart valves (esp. **mitral**) over time

Valvular disorders (mitral/aortic)

- **Stenosis** = valve too **narrow/stiff** → the chamber behind it works harder to push blood through → pressure overload/hypertrophy.
- **Regurgitation (insufficiency)** = valve **leaks backward** → blood flows the wrong way → volume overload.
- Net effect: reduced forward output, murmurs, and eventually **heart failure**.

□Memory hooks

Layers: **Peri**carditis (sac, pain better leaning forward), **Myo**carditis (muscle, weak pump), **Endo**carditis (valves, vegetations). **Tamponade = Beck's triad** (hypotension + JVD + muffled sounds). **Rheumatic = after strep → mitral valve. Stenosis = stiff/narrow; regurg = leaky.**

⚠ **Common confusion**

Pericardial effusion vs tamponade: an effusion is just fluid; **tamponade** is when that fluid **compresses the heart** and drops cardiac output — an emergency. Pericarditis pain **improves** leaning forward (vs MI pain).

Practice Questions

1. A patient has hypotension, JVD, and muffled heart sounds after a large pericardial effusion. What is this triad and diagnosis?

   Show answer

   Beck's triad → cardiac tamponade. Fluid compresses the heart so it can't fill; cardiac output falls — an emergency treated with pericardiocentesis.

2. A patient has sharp chest pain relieved by leaning forward and a pericardial friction rub. Which condition?

   Show answer

   Pericarditis. The positional pain (better sitting forward) and friction rub are classic — and help distinguish it from MI.

3. Untreated streptococcal pharyngitis can lead to which chronic valvular disease, and which valve is most affected?

   Show answer

   Rheumatic heart disease — an autoimmune reaction after strep scars the valves, most commonly the **mitral** valve.

Clinical scenario

A person who injects drugs develops fever, a new heart murmur, and small painful nodules on the fingers with splinter hemorrhages under the nails. **What is the likely diagnosis and mechanism?**

Infective endocarditis *— bacteria seed the valves and form* ***vegetations****; pieces embolize peripherally (Osler nodes, Janeway lesions, splinter hemorrhages). IV drug use is a major risk factor.*

TOPIC 9

□Peripheral Vascular Disease

HIGH YIELD ▼

Vascular disease outside the heart: arteries (poor inflow), veins (poor outflow), and vasospasm.

Condition	Patho & signs
Peripheral artery disease (PAD)	**Atherosclerosis** narrows leg arteries → poor **arterial inflow**. Signs: **intermittent claudication**, cool/pale/hairless legs, weak pulses, pain worse with elevation, wounds that don't heal (arterial ulcers on toes/feet)
Intermittent claudication	The hallmark PAD symptom: **cramping leg/calf pain with walking that is relieved by rest** (muscle ischemia when demand rises)
Venous insufficiency	**Incompetent valves** → poor **venous return** → blood pools. Signs: **edema, brown/hyperpigmented skin, venous stasis ulcers (medial ankle)**; complications include chronic ulcers and DVT risk
Raynaud's	**Vasospasm** of small arteries (triggered by **cold/stress**) → fingers/toes turn **white** → **blue** → **red** with numbness, then rewarming

□Memory hooks

PAD = arterial = pain with walking (claudication), cool/pale, distal ulcers. Venous = pooling = edema, brown skin, ankle ulcers. Raynaud's = white-blue-red with cold/stress.

⚠ Common confusion

Arterial (PAD) vs venous: arterial = **poor inflow** (cool, pale, pain with activity, distal/toe ulcers); venous = **poor outflow** (warm, edematous, brown skin, ankle ulcers). Opposite pictures.

□Practice Questions

1. A patient reports cramping calf pain when walking that stops after resting a few minutes. Which sign, and which disease?

Show answer

Intermittent claudication → peripheral artery disease. Narrowed arteries can't meet the higher oxygen demand of walking, so the calf becomes ischemic until rest lowers demand.

2. A patient's fingers turn white, then blue, then red with numbness when exposed to cold. Which condition?

   Show answer

   Raynaud's — cold/stress-triggered vasospasm of the digital arteries produces the classic white→blue→red color change.

3. **SATA:** Which suggest chronic venous insufficiency? (a) lower-leg edema, (b) brown/hyperpigmented skin, (c) ulcers at the medial ankle, (d) cool pale foot with absent pulses, (e) pain with walking relieved by rest.

   Show answer

   a, b, c. Edema, brown skin, and medial-ankle ulcers are venous. (d) and (e) are **arterial** (PAD) findings.

Clinical scenario

A diabetic smoker has a cool, pale, hairless lower leg with weak pulses and a non-healing ulcer on the toe. **Arterial or venous, and why?**

Arterial (PAD) *— atherosclerosis limits inflow, so the tissue is cool, pale, and hairless with weak pulses and poor healing; ulcers form distally (toes/feet). Smoking and diabetes are major risk factors.*

TOPIC 10

□Congenital Heart Defects

HIGH YIELD ▼

Congenital heart defects are structural problems present at birth. The key split is **cyanotic vs non-cyanotic** — based on whether deoxygenated blood reaches the systemic circulation.

	Non-cyanotic (acyanotic)	Cyanotic
Shunt	**Left → right** (oxygenated blood recirculates through the lungs)	**Right → left** (deoxygenated blood enters the body)
Cyanosis?	**No** (blood is still oxygenated)	**Yes** (bluish skin)
Examples	**VSD, ASD, PDA**, coarctation of the aorta	**Tetralogy of Fallot**, transposition of the great arteries

Tetralogy of Fallot (the classic cyanotic defect)

Four defects (mnemonic **PROVe**): **P**ulmonary stenosis, **R**ight ventricular hypertrophy, **O**verriding aorta, **V**entricular septal defect. Because pulmonary stenosis + the VSD push deoxygenated blood out to the body, the child is **cyanotic**. Signs: **cyanosis, "tet spells"** (worsening cyanosis with crying/feeding), and a child who may **squat** to relieve a spell.

□Memory hooks

Left→Right = no cyanosis (oxygenated blood recirculates): VSD, ASD, PDA.
Right→Left = cyanosis (deoxygenated blood to body): **Tetralogy of Fallot**, transposition. ToF = **PROVe** (Pulmonary stenosis, RVH, Overriding aorta, VSD).

⚠ Common confusion

Direction of the shunt sets the color: **L→R shunts do NOT cause cyanosis** (blood re-runs through the lungs), while **R→L shunts cause cyanosis** (deoxygenated blood bypasses the lungs into the body).

□Practice Questions

1. Which type of shunt causes cyanosis, and give one example defect.

Show answer

A **right-to-left shunt** causes cyanosis because deoxygenated blood enters the systemic circulation. Example: **Tetralogy of Fallot** (or transposition of the great arteries).

2. What are the four components of Tetralogy of Fallot?

   Show answer

   PROVe: Pulmonary stenosis, **R**ight ventricular hypertrophy, **O**verriding aorta, **V**entricular septal defect.

3. **SATA:** Which are non-cyanotic (left-to-right) defects? (a) VSD, (b) ASD, (c) Tetralogy of Fallot, (d) PDA, (e) transposition of the great arteries.

   Show answer

   a, b, d. VSD, ASD, and PDA are non-cyanotic (left-to-right). Tetralogy of Fallot and transposition are cyanotic (right-to-left).

Clinical scenario

An infant becomes deeply cyanotic during crying and feeding and instinctively squats. **What defect, and why does squatting help?**

Tetralogy of Fallot *("tet spells"). Squatting raises systemic vascular resistance, which reduces the right-to-left shunt and pushes more blood through the lungs — temporarily improving oxygenation.*

Exam Cheat Sheet — Night-Before Review

Module 9 — Blood Disorders

- **Anemia s/s (all types):** fatigue, pallor, tachycardia, dyspnea on exertion, dizziness.
- **Iron deficiency** = microcytic, ↓**ferritin**, spoon nails, pica. **Thalassemia** = microcytic, **normal iron**, inherited, target cells.
- **B12/folate** = macrocytic (DNA defect); **only B12 → neuro signs**. **Pernicious anemia** = autoimmune loss of intrinsic factor → no B12 absorption.

- **Sickle cell** = HbS sickles when deoxygenated → vaso-occlusion (triggers: hypoxia, dehydration, cold, infection). **Aplastic** = marrow failure → pancytopenia.
- **Von Willebrand** = ↓vWF → mucosal bleeding, normal platelets, long bleeding time. **DIC** = clots then bleeds; ↑PT/aPTT, ↓platelets/fibrinogen, ↑↑D-dimer.
- **Polycythemia vera** = JAK2, too many RBCs, ruddy, itchy after warm shower, **clot risk**. **Acute hemolytic transfusion reaction** = ABO mismatch → STOP + saline.

Module 10 — Cardiovascular

- **Atherosclerosis:** injury → fatty streak (foam cells) → fibrous plaque → **rupture + clot** = acute MI. CAD modifiable risks: smoking, HTN, high LDL, diabetes.
- **Angina/MI:** stable = exertion, relieved by rest. Unstable = rest, **troponin normal**. NSTEMI = ↑troponin, no ST elevation. STEMI = ↑troponin + **ST elevation** (emergency). **Troponin** = cardiac injury marker. **Sudden cardiac death** = VF from CAD.
- **Heart failure:** Left = Lungs (dyspnea, crackles); Right = Rest of body (JVD, edema). Systolic = weak squeeze (↓EF); diastolic = stiff, can't fill (EF preserved). **#1 cause of right HF = left HF. BNP up = HF.**
- **Preload** = filling volume/stretch; **afterload** = resistance to eject (↑ by HTN). **ANP/BNP** = released when heart is stretched → excrete Na/water, vasodilate.
- **Hypertension** = silent; damages heart, brain (stroke), kidney, eyes. **Blood-loss compensation:** sympathetic (fast) → RAAS/ADH → EPO (slow).
- **AFib** = quivering atria → clot → **stroke** (anticoagulate). **VFib** = cardiac arrest → defibrillate.
- **Pericarditis** = pain better leaning forward + rub. **Tamponade** = Beck's triad (hypotension, JVD, muffled sounds). **Myocarditis** = viral, weak pump. **Endocarditis** = valve vegetations (IV drug use). **Rheumatic** = post-strep → mitral valve.
- **PAD** = arterial, claudication, cool/pale, toe ulcers. **Venous insufficiency** = edema, brown skin, ankle ulcers. **Raynaud's** = white-blue-red with cold.
- **Congenital:** L→R = no cyanosis (VSD, ASD, PDA); R→L = cyanotic (Tetralogy of Fallot = PROVe, transposition).

□Don't-Mix-These-Up

Easily confused	The key difference
Iron deficiency vs thalassemia	Both microcytic — iron def = **low ferritin**; thalassemia = **normal iron**, inherited.

B12 vs folate	Both macrocytic — only **B12** causes neurologic signs.
Unstable angina vs NSTEMI	Both rest pain, no ST elevation — NSTEMI has **positive troponin** (cell death).
NSTEMI vs STEMI	STEMI = **ST elevation + complete occlusion** (emergency reperfusion).
Systolic vs diastolic HF	Systolic = weak squeeze (↓EF); diastolic = stiff, can't fill (EF preserved).
Left vs right HF	Left = **Lungs** (crackles, dyspnea); Right = **Rest of body** (JVD, edema).
Preload vs afterload	Preload = volume coming in (stretch); afterload = resistance to eject (HTN raises it).
AFib vs VFib	AFib → **stroke** risk (anticoagulate); VFib → **cardiac arrest** (defibrillate).
Effusion vs tamponade	Effusion = fluid; tamponade = fluid **compressing the heart** (Beck's triad) — emergency.
PAD (arterial) vs venous insufficiency	Arterial = cool/pale, claudication, toe ulcers; venous = edema, brown skin, ankle ulcers.
Cyanotic vs non-cyanotic defects	L→R = **no cyanosis** (VSD/ASD/PDA); R→L = **cyanotic** (ToF, transposition).

□Glossary

Anemia — reduced oxygen-carrying capacity (low Hb/RBCs).

MCV — average red-cell size; classifies anemia as micro-, normo-, or macrocytic.

Ferritin — stored iron; low in iron-deficiency anemia.

Intrinsic factor — gastric protein needed to absorb B12; lost in pernicious anemia.

von Willebrand factor — protein that helps platelets adhere; deficient in vWD.

DIC — disseminated intravascular coagulation: widespread clotting then bleeding.

Polycythemia vera — JAK2-driven overproduction of RBCs → thick blood, clot risk.

Atherosclerosis — lipid plaque buildup in artery walls.

Troponin — protein released with myocardial injury; elevated in MI/myocarditis.

STEMI / NSTEMI — complete vs partial coronary occlusion causing infarction.

Ejection fraction — % of blood ejected per beat; reduced in systolic HF.

Preload / Afterload — ventricular filling volume / resistance to ejection.

BNP — peptide released by stretched heart; elevated in heart failure.

Atrial fibrillation — disorganized atrial activity → clot/stroke risk.

Cardiac tamponade — pericardial fluid compressing the heart (Beck's triad).

Intermittent claudication — exertional leg pain relieved by rest (PAD).

Tetralogy of Fallot — cyanotic defect: pulmonary stenosis, RVH, overriding aorta, VSD.

□Final Study Tips

- **Blood: anchor every anemia to its lab clue** — MCV size (micro/normo/macro), ferritin, and the one hallmark (spoon nails, neuro signs, sickled cells).
- **Heart: cause → effect.** Ask "forward or backward failure?", "supply or demand ischemia?", "left or right?" and the signs follow.
- **Lock in the opposite pairs:** stable vs unstable angina, NSTEMI vs STEMI, systolic vs diastolic HF, left vs right HF, cyanotic vs non-cyanotic defects, preload vs afterload.
- **Know the emergencies:** STEMI, cardiac tamponade (Beck's triad), DIC, acute hemolytic transfusion reaction.
- **Troponin** = the cardiac-injury marker (rises in MI/myocarditis). **BNP** = the heart-failure/volume marker.
- **Practice SATA carefully** — read each option as its own true/false statement.
- **Use the cheat sheet** the night before AND the morning of the exam, then self-test with answers hidden. Format: 50 questions, Respondus LockDown Browser.

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You've got this. Master each disorder's one hallmark, and the scenarios practically answer themselves.

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